

Game Audio Explained – your guide to making sound for games

[Jennifer Walden](#) January 29, 2025



Welcome to our brand new 'Sound Explained' series where we take you on a definitive deep-dive into common audio terms and concepts - like Game Audio. But, unlike a regular Wikipedia post where anyone can contribute, this one is written and reviewed by some of the sharpest minds and ears in game audio.

The idea is to give you distilled key insights and knowledge about everything game audio, and provide you with the best jumping-off points for further exploration.

In this guide, you'll learn about everything from the fundamentals of game audio and how it's done from start to finish, game audio roles, important considerations and key game audio tools, to middleware, sound recording, how to learn game audio + how to get started and succeed in game audio - and tons more.

Hope you find it useful - if so, please share it in your network so more people can benefit from it. And without further ado, let's dive into the fantastic world of game audio!

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What is game audio?

Game audio is the broad term for the art of sound in interactive media. It's an umbrella term that brings together the creative, technical, and collaborative aspects of sound design, dialogue, music, and mixing with the multidisciplinary world of game software development.

Video game development unites various disciplines, such as coding, visual design, animation, sound, lighting, production, game design, and narrative. As such, they create a multisensory experience through the synergy of these individual components, amplifying each other. This means that as a game audio specialist, you'll be part of a highly diverse team, creating something unique through the combined efforts of everyone involved.

GAME AUDIO AS AN INVISIBLE ART

Game Audio is an invisible art – both music, voice, sound and mixing are perceived on an auditory level. When done right, it can ground the player in the experience, and makes fantastic and fictional worlds feel real. Audio can also be a passive experience, meaning – we don't actively choose what we listen to, but are guided by the mix. Because of that, sound is a great vehicle into our subconscious mind, it influences players' feelings before they even have a chance to process what just happened.

Game Audio can be used in powerful ways to change a player's perception of a game – whether it would be the powerful feeling of a magic spell, the suspenseful feeling we get from a creaking door or the mysterious mood of a forest set by the music. When Game Audio is created, designed, implemented and mixed well, players can be fully immersed in the gameplay and story experience, forgetting that the game was intentionally designed to make us feel a certain way by a team of designers.

THE DIFFERENCE BETWEEN GAME AUDIO AND LINEAR AUDIO

Perhaps the most fundamental difference between game audio and linear audio for film and TV is that game audio is not a 'post-production' practice; meaning that in game audio, we do not work exclusively in a post-production environment to (near) complete visuals.

| *Game audio is an iterative craft; meaning that game audio work is done earlier and is more integrated into the development process*

Game audio is an iterative craft; meaning that game audio work is done earlier and is more integrated into the development process – making continual changes as the game itself changes, and keeping up with any shifts in design or production. In this sense, it is much more of a collaborative craft, and much more of a craft of discovery along with the development team. This is not to say there are no 'post-production' aspects to game audio, there are also many of those too, from scoring or designing sound for cinematic cut-scenes, to music recording and orchestration, through to the very late final mixing process. However, broadly speaking, the craft is considered an iterative one rather than post-production one.

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GAME AUDIO ROLES

What are some of the roles on the game audio team?

THE FOUR FIELDS OF GAME AUDIO

Broadly speaking, there are four main development fields in game audio: Music, Sound Design, Dialogue, and Integration (though *integration* is increasingly a part of the other three areas, rather than a separate field). Integration is implementing the audio in a game, typically using a middleware like [Wwise](#) or [FMOD](#) but sometimes going as far as visual scripting and coding bespoke systems directly within the game's engine.

Smaller projects are more likely to have a very small number of audio professionals with a broader skill set involved

The number and titles of positions in the audio department vary significantly depending on the studio, project, team size, and budget. Most game audio professionals know one field well and perform the rest of them if needed* (see generalists vs specialists). Smaller projects are more likely to have a very small number of audio professionals with a broader skill set involved, and would usually only contact a game audio professional around the end of the production, while larger projects require a much bigger

team, therefore these fields are often divided into smaller roles, allowing people to specialize in specific areas of expertise. Additionally, many studios rely on external companies that focus on one or more of these fields.

The increasing prevalence of audio outsourcing and post-production companies also means that a game studio might not even have an audio department but has outsourced the entire pipeline to a third-party instead. Consider that training and providing facilities for an in-house audio team can be a costly and time-consuming affair, especially for a studio that only recently received funding or finds itself in a phase of rapid growth.

GAME MUSIC

Music is probably the category in game audio that players are the most consciously aware of. This most often refers to the composed score that you hear in a video game, and is usually meant to set the emotional tone and the feel of the game world and story. Some composers will write music for video games in what could be considered a traditional linear way, closer to the way most music scores for film or TV is being made. This method is therefore easier for musicians since it resembles the process of any music creation. The score is then edited and implemented by a separate integration team in order for it to feel reactive to whatever the player is doing in the game.

An interactive music system [...] refers to music that is written and designed from conception to be adaptive to the game

The other increasingly common approach for composers is an interactive music system. This term refers to music that is written and designed from conception to be adaptive to the game and changes based on certain parameters from the game engine – e.g., when the player moves to a new location, when the player's health level goes below a certain threshold, when the player encounters combat, and so on. In between or on top of longer musical segments, composers can add 'stingers.' A stinger is a one-

shot musical cue that is triggered at a specific moment, or event. (For example, you might hear a stinger when the player encounters an enemy). A musical stinger can be triggered 'on the beat' within the musical tempo and rhythm of the music that may be currently playing.

GAME SOUND DESIGN

Sound design is a vast category, and includes many sub-categories like environmental ambience sound, physics collision sounds, menu UI (user interface) sound, weapons, vehicles, character sounds such as footsteps, cloth or gear movement (foley), in-game notifications (reward or failure sounds, as well as collectables and pickups), and so on. When designing sound, dividing sound assets into categories based on their playback in a game can be useful. For example, 'one-shots' refer to any sound that is triggered once (or if triggered more than once, accounting for randomized variations), and 'loops' refer to sounds that keep playing in a loop, but that also may contain variations such as water lapping.

Dividing sound assets into categories based on their playback in a game can be useful.

Some sound designers work almost exclusively in a DAW (digital audio workstation) such as [REAPER](#), [Pro Tools](#) or [Nuendo](#) to create and export their assets, later transferring them to the integration/implementation specialist.

More commonly sound designers in interactive entertainment are expected to create and integrate their sound into the game engine.

DIALOGUE

Dialogue is the process of recording voice overs for characters in the game. It can be something simple like instructions in a tutorial but can easily get really complex when you have a story driven game with many characters. In some games, vocal performance can also include voice

effort sounds (e.g., grunts and screams from getting hit with or swinging a sword) and 'barks.' 'Barks' refers to random one-liners from NPCs, for example: "What are you buying?" and "I used to be an adventurer like you. Then I took an arrow in the knee..."

If the game is intended to be sold in multiple countries, every line would have to be translated to all of the supported languages

A major challenge in dialogue creation is revisions. If a text was recorded but the game team suddenly decided to change a line or add a word, they will need to bring in the same voice actor/actress and record in the same exact conditions to ensure that everything sounds coherent.

Another challenge in dialogue has to do with localization. If the game is intended to be sold in multiple countries, every line would have to be translated to all of the supported languages.

Because of these complexities, often studios will partner with an external company for all their dialogue needs, or use AI voices to make things easier.

IMPLEMENTATION

An implementation or integration specialist is the person who works closely with the engineering team, ensuring that all of the audio assets (i.e., music, sound design, and dialogue) are integrated into and hooked up to cues in the game. Implementation specialists can come from a engineering background rather than a sound or music background. They are the bridge between the audio and the game. It's important to understand that almost nothing in game development happens automatically, and connections between game events and their audio counterparts all have to be created by someone.

The job of an audio implementation specialist can be straightforward, meaning they receive a sound from the sound designer, the sound needs to play every time the player hits a trigger (for example, every time a coin is

collected), and the implementation specialist connects the trigger and the sound that should be played.

Other times, they have more freedom to determine how they want things to sound. For example, they will receive a list of 10 sword swishes from the sound designer, and will have to create an audio system where every time the player swings a sword, one out of the 10 sounds would play randomly, in different volumes. So instead of simply implementing the sound, they will make sure the sound is engaging and not too repetitive.

Sounds can also be created procedurally at runtime, for example through the use of synthesizers – or the entire musical arrangement is created procedurally through MIDI – which then plays back sampled or procedural instruments. The use of parameters allows Technical Sound Designers to adjust audio through information that is coming from the game engine. This can be used effectively to emphasize elements in the entire sound mix that are important to the player, for example.

THE TECHNICAL SOUND DESIGNER

A common title for someone combining sound design skills with integration skills is “technical sound designer.” Over the years, this role has become more prevalent. Technical sound designers are individuals who have a strong creative but also technical background, and can carry through a design task with technical understanding. The roles and tasks for this title can vary from studio to studio. Generally speaking, a technical sound designer works “close to the game engine.” That means their work ranges from authoring playback structures for audio middleware (like [Wwise](#) or [FMOD](#) – sometimes known as ‘the audio engine’) to audio scripting and programming in the game engine.

Generally speaking, a technical sound designer works ‘close to the game engine.’

Luckily (or unfortunately to some people), no two jobs are alike. Two

technical sound designers from different studios could have very different roles and responsibilities. The requirements can range from designing audio systems from scratch in the company's chosen engine or programming language to working in middleware and scripting within the game engine. So, based on the job description, it makes sense to ask any studio what exactly they are looking for in their technical sound designer role. If technical sound design is something you aspire to pursue, pay close attention to job postings and their requirements, while keeping in mind that the role will evolve and change as the project evolves.

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GENERALISTS, SPECIALISTS, AND T-SHAPED

Some roles in game audio are considered specialist, meaning the focus of the work, and the responsibility is on one singular aspect of some of the work described above, such as an integration specialist. Other types of roles within game audio are considered generalist, meaning that a deep specialization isn't needed in the work, but a general understanding of how to do everything is necessary, as the work involves jumping around from one area to another in order to get audio into the game.

A more fitting description of the work of a game audio developer, however, is somewhere in between. Sometimes referred to as a T-Shaped person, this is someone who has very wide generalist skills (the broad horizontal line at the top of the letter 'T'), and can move around within the team and do a lot of different things, combined with a deep specialization, for example in weapon sound design, or in foley (represented by the vertical line of the letter 'T').

Some audio teams are made up of a mix of specialists, generalists, and T-shaped people, depending entirely on how a specific team works to make a

specific game. On a very small indie team for example, an audio person may find themselves writing the music, designing the sound effects, and integrating everything into the game engine, as well as collaborating very closely with the rest of the small team and even suggesting game design mechanics across disciplinary borders. On a large AAA team, an audio designer may find themselves inside a very tightly defined hierarchy with specific responsibilities, with other members of the team covering other specific aspects of the sound's creation.

As game audio people move from project to project throughout their careers, they are likely to encounter all of the above and more, and as they experience different ways of working, they may discover specializations as well as learn generalist skills. Over time, specialists tend to move more towards generalists, and generalists tend to develop one or more specializations, eventually arriving at the T-shaped skill set.

When working in game audio, it's also important to be aware of how you and your work relates to other developers on the broader team. Ashton Mills has written a great overview of the game development structure, roles, and how game audio relates to these roles [here](#) on A Sound Effect. Also, check out Alex Haisting's articles on "Getting Into Game Audio: 8 Typical Roles and Career Paths" [here](#) on Waveinformer.

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How is game audio created?

As mentioned previously, there is no one way of doing things. The workflow can change based on the team size, game style, studio culture, budget, and so on. Practically speaking, there are a few stages that occur commonly across most video game development cycles: establishing a feel, spotting session, asset creation, integration, and mixing.

ESTABLISHING THE FEEL/TONE

Establishing a feel is the first and probably most crucial step to creating a game. In this phase, the audio director, or sometimes audio lead, works with the broader team (e.g., art director, creative director, etc.) on articulating a shared vision of what it is the team wants to create. In this phase, there can be a lot of experimentation and prototyping, trying to zero-in a coherent idea of how the team wants the game to sound and feel. Oftentimes, it would be on the audio director or lead to define the vision, and prepare examples and presentations of the core ideas that are easy for anyone to understand, whether they are on the audio team, or whether

they are an artist or animator.

Once the vision is established, usually in the form of 'audio pillars,' [...] it is then a matter of getting examples on-screen

Once the vision is established, usually in the form of 'audio pillars,' and communicated via a [game audio design document](#), it is then a matter of getting examples on-screen. This is often done through prototyping particular features of the game with the rest of the team, and trying out the audio vision to make sure it is working, and making any tweaks as necessary.

Once the tone is established, the rest of production can move forward, using the pillars and the tone guides created in the concept and pre-production phase, against which to evaluate the work.

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THE SCOPING (or 'spotting') SESSION

After establishing the creative direction, sometimes the next step can be a scoping/spotting session (a concept borrowed from post-production sound approaches). In this phase, the team or the individual person in charge of each element of audio would go over the entire game together with a game director and estimate all of the moments that require different kinds of sounds, music cues, or mix moments.

All of the information is gathered in a document, specifying any details of the sounds

All of the information is gathered in a document, specifying any details of the sounds. It could include information like: how many variations, what

should be the ideal length of each sound, how the sounds could be played (one shot loop, randomly triggered) and so on. It could also include production notes (such as 'acquire sound library,' or 'record sounds/foley'). For example, the initial list can include something like: "estimate we'd need ~10 one-shot sword swishes and five one-shot sword impact sounds for the second scene." Experienced sound designers develop templates that they use for each similar project they work on, gathering all of that information in one place.

This process can give the team a sense of the SCOPE of the work involved in sound, music, and voice asset creation, and in that sense can be crucial in determining how much time and cost would be associated with a project's current scope.

Note: Scope can change dramatically throughout the development of a game, with features and elements being cut or added as the game is created. This process tends to occur earlier in the game development process. However, almost all games still accommodate drastic changes even during the later production periods.

THE PRODUCTION PHASE

The production phase is where all the assets are being sourced or created. Sound designers can use a variety of ways to create assets, such as field recording, studio recording, generating sounds using synthesis, or mixing several sound elements together to create a new composite sound. Eventually they will all be arranged in a DAW (digital audio workstation) and exported based roughly on the sheet that was created during the scoping/spotting session. Increasingly, integration into the audio engine, or middleware, is considered a part of the production process, but we will discuss this step separately next.

THE IMPLEMENTATION PHASE

The next phase is implementation, and can be thought of as the secondary part of production. In this phase the technical sound designer or audio implementer would take all of the created assets and start implementing them into the game engine. Sometimes teams use a middleware like [Wwise](#) or [FMOD](#) to implement the sounds, other times it would be directly implemented into the game engine using perhaps [Unreal Engine's MetaSounds](#) framework.

The implementation phase can also include the process of creating audio systems

This phase is the first time that the sound would be triggered and heard in context. Using our example from before – the 10 recordings of sword swishes – they would be hooked up together to hear how they would sound when the player hit the attack button to swing the sword. This phase may require some iteration between the sound designer and the technical sound designer to get everything to sound right (another reason it is becoming more common for sound designers to implement their own sounds). Sometimes the ideas from the DAW don't translate as well as was thought to the game engine and need a few revisions to sound as intended.

The implementation phase can also include the process of creating audio systems. These systems can be for the music that is playing in the game, the dialogue sentences and lines being triggered, or individual systems for sounds within the game such as physics collisions or 3D ambience spot-effects or reverb volumes.

Eventually, all of the sounds for the game need to be incorporated into the game engine and played based on the user interactions or navigation. The rules can vary between teams and type of game but the overall process is generally quite similar for all of them. The integration process is considered 'complete' when all the sounds are hooked into correct triggers and behaviors in the game, and all the sounds have been thoroughly reviewed and approved as finished.

For more, check on Jaime Cross's [A Guide to Game Audio Task Planning](#).

MIXING AND BALANCING

The last phase is mixing and balancing all the audio in the game. This phase may be broken down into pre-mixing and final mixing. Pre-mixing is where each food group of audio: sound effects, dialogue/voice, foley, and music, are leveled to their specified target loudness. These food groups are then mixed against each other in context to make sure they coexist well with one another.

Final mixing is often followed by a mastering phase, in which the sound output for different game platforms is carefully tweaked and tuned

The final mix is also the moment where major decisions can be made about what is the most important thing for the player to hear, and often this is achieved by ducking other sounds out of the way in order to achieve clarity at specific moments in gameplay. In this phase, the mixing team might also adjust things like in-game reverb levels, EQ and filters per sound or per sound group. That will help all the sounds feel like they are part of the same environment. This phase is necessary to make the overall game feel polished, satisfying, and coherent for the player and can take anywhere between one to five weeks, depending on the scope of the game.

Final mixing is often followed by a mastering phase, in which the sound output for different game platforms is carefully tweaked and tuned, based off of the master mix, to achieve parity between different output and platform formats.

To learn more about game audio mixing, check out Audio Director Alex Riviere's guide, [Demystifying Game Audio Mixing](#).

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How does creating sound compare across different types of games – indie, AAA, mobile, etc.?

The type, style, genre, and platform of a game have a huge influence on the game development process as a whole, and game audio is also included.

One of the key differences between platforms is the amount of memory that can be allocated for a single game.

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The genre of a game is crucial to developing a sonic palette and a strategy for the game's sound. This is where references to other games and media come in really handy – both for finding references of what the game should sound like and what the game should not sound like. Next-gen productions often task themselves with pushing the envelope and working with technologies and principles of the future. This includes a high degree of experimentation. So there is no 'one size fits all' solution here. It's all about

teamwork, dedication, and exploration in order to find the strategy and direction that suits your project the best.

MOBILE GAME AUDIO

Many mobile games in the past did not utilize middleware in order to keep production costs low and stay within tight hardware resource budgets, though this is changing as production values get higher in the mobile market. It is not uncommon for sound designers and composers to solely work as asset creators on these kinds of games. While knowing the ins and outs of the game engine's native audio tools can be a benefit here (e.g., [Unreal Engine's MetaSounds](#), [Unity Audio](#), etc.), it is more likely that the programmers working on these titles will want to implement audio assets themselves as there tends to be far fewer of them than in a AAA game for comparison.

INDIE GAME AUDIO

Indie development teams tend to be small, as are their budgets at times. As such, they often only have the budget for a single audio person (see generalists) to work on a project. It is quite common to see people handle all audio tasks from sound design, to composition, to implementation throughout the game's production process.

Whether or not you will have the benefit of working with middleware will depend on the studio's budget, production time, and willingness to embed third-party tools to get more frictionless integration.

AAA GAME AUDIO

In AAA games, audio roles tend to be more specialized (see specialists and T-shaped) as it's much more common to see audio teams consisting of roles each with different responsibilities. While many company's in-house audio teams still tend to be quite small, there are rare examples of teams of

50 or more audio people working on games – like [God of War: Ragnarök](#) – at different times during production.

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What are some important considerations when working on sound for a game?

There are many important considerations that are taken into account when working on sound for games. Those considerations might feel like constraints for some creators, but they can also be viewed as guidelines, making it easier to stay focused on the goal.

PREPARATION

Preparation is key to a successful audio experience. Therefore, when joining a project, it's crucial to establish a clear understanding of the game's tone early on. Gathering as much information as possible about the game's story, its inspirations from other games, movies, or shows, and accessing the game design document, if available, can provide valuable

context. This knowledge informs stylistic decisions and offers insights into the source material needed. Additionally, it allows for important budget considerations, such as whether additional libraries, field recording sessions, or specific tools will be required to achieve the game's full potential.

BUDGET

Speaking of budget consideration, it's always important to keep realistic expectations when it comes to time and effort. Some projects would have a limited budget, but will ask for a lot more than they can afford. It's not because they are trying to scam creators; usually the broader team just don't know how to assess the work that is required for good game audio. Some jobs seem easy but take a long time. Others sound very complex but are fairly trivial to a professional creator. So, staying aware that the time and effort estimation is on the musician/sound designer, and not on the studio is key to avoiding exploitation and burnouts.

RESOURCES

Besides the financial budget, an important thing to keep in mind is the memory budget. As all code, animations, graphics, and sound compete for CPU and RAM resources in real-time, sound often has to dial back on the resources available. Games can get pretty big in size (some games get to 120GB!), but depending on the platform they are running on, there can be some limitations on how much data and memory can be used for a single game.

Mobile devices (such as phones and XR headsets) have a limited capacity. Considering the player often has more than one game on the system at any given time, the size of the games is limited to a maximum of 5GB. This especially influences the amount of audio compression necessary, as well as some other optimization choices including the amount of variations that can be played for an individual sound event, and the length of any sound loop. Sound files can be set to stream from your hard drive, which puts less

strain on your memory but can introduce some latency in playback – this option is not compatible with sound effects that have to trigger immediately.

With AR/VR games, there's an added challenge to achieve a high enough framerate (closer to 90 frames per second) in order to avoid motion sickness for the players.

Platforms like PC and console could afford more memory and significantly more compute power, allowing game audio professionals to work with much more granular approaches for sound design, like using more variations or layering different elements together.

THE DEVELOPMENT WORKFLOW

Another consideration, particularly with less experienced developers, is to plan when each part of the audio can be worked on. For example, animation-dependent sound effects can only be created when the animation is (mostly) done, while composing music (or at least looking for reference tracks) can be done while the visual and narrative departments are still working on style and story. Nevertheless, it is a great idea to explore sound design even in the early stages of game development and to develop a sonic palette, which serves as a blueprint for your sound design in terms of style and references.

It is a great idea to explore sound design even in the early stages of game development and to develop a sonic palette

At the end of the day, each role in game audio has to make some decisions that will establish the approach, and eventually the overall result of how the experience would sound. If there's an audio director on the team, most of the decisions would be deferred to the director, and the team would use the provided guidelines to work on the game. If the project is much smaller, and the team can only afford one person for all of the sound in a game, some parts will take priority over others (e.g., if the person is a technical

sound designer, they might choose to focus on the sound design and implementation aspect, and use libraries or royalty free music instead of composing everything themselves).

Considering the different fields and the different structures of companies, each role has some decisions to make in order to create a meaningful experience – decisions are typically taking into account creative ideas and can be impacted by budget and compute limitations.

GAME MUSIC CONSIDERATIONS

For music, the main question is whether or not to include real instruments/real musicians and to what degree. This is obviously a very creative choice and depends on the style of the game. For example, sci-fi games could both use a synthetic score or – for the sake of contrast – go fully orchestral.

If using musicians, the musicians would need to arrange a score/find a way to communicate their musical idea, and set up a recording session. For music created in a DAW (digital audio workstation), more time is needed for designing unique sounds and automating them to make them sound “alive.”

GAME SOUND DESIGN CONSIDERATIONS

The sound designer(s) have to ask themselves to what degree they can and want to rely on sound libraries or record new source material. This step can range from foley work to even completely new vehicle recordings – fully depending on the project. Generally, most sound designs are created when the visual animation is at least roughly finished, so the sound really matches the visuals of the game.

DIALOGUE CONSIDERATIONS

The dialogue team has to be in close contact with the narrative team to

have a good idea of how the characters should sound and behave, to guide the dialogue recordings with the voice actors accordingly. Using AI voices for NPCs is becoming more common, as various voice characteristics are easily accessible and testable. However, by using artificial voices you lose the value of having a directable living voice actor who can bring authenticity to your cast.

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In addition, it is best to keep up to date on valid concerns by the voice acting community, such as how best to protect their rights in regards to fair pay if recordings are used for AI and/or how best to protect your cast in their contracts to guarantee that their voice recordings will NOT be used for AI and synth learning. This is a very important topic for the past two years and should not be dismissed lightly. Following resources like [NAVA](#), learning about their [fAIrvoices project](#) and the now highly-recommended and widely used [NAVA AI Rider](#) with your voice actor contracts, and staying up to date on why [SAG AFTRA Interactive \(that includes games\) actors went on strike twice](#) in two years to fight for voice protections is highly recommended.

GAME INTEGRATION CONSIDERATIONS

Finally, the integration team is normally closely connected with all other audio team members and the development team; they are the glue that brings everything together. As all the audio comes together “on their desk,” they will have one of the first impressions of multiple elements combined. This means they are in charge of mixing the sounds and music but also sometimes have to ask for reworks, as some material just won’t work together.

One of their considerations is how to mix all the sounds into one coherent game. In terms of spatialization, they have to determine which sounds can be mono, which ones are stereo, and which are spatial. Depending on the

project, the development team or an additional audio programmer may even take care of the coding aspect in the game engine, or develop tools for members of the audio team, to improve and streamline workflow pipelines.

All the steps listed above can potentially be split up among different people, which is why a very good organization and documentation is crucial. For all of these disciplines, it is very important to stay in close contact with the different audio departments and the director/producer of the game to regularly ensure the progress is going in the direction that the entire game team was looking for.

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What is game audio localization?

Localization is the process of translating the game experience to different countries. The most straightforward part of localization is translation of text, but it can be so much more. It's a cross-functional effort that can be handled by numerous teams and collaborators to adapt a game for

different regions. This process can have wide-ranging implications. Sometimes the visuals need to be adapted, signage and textures need to change, characters need to be modified, and, of course, all the dialogue and in-game text have to be translated and adapted.

The cornerstone of any successful localization project is getting the source script right

If the game needs to support languages that are written in different ways (for example, German, Arabic, and Chinese are all written differently) that might create another challenge for the designers. In some games, they took it as far as creating different characteristics or different items based on the country they sold it in. A great example is [Animal Crossing](#). Although all the characters speak gibberish, the studio made an adaptation for how the gibberish sounds, for instance, in the Japanese version of the game.

The cornerstone of any successful localization project is [getting the source script right](#). To ensure this, a thoroughly proof-read source script with consistent term usage is a must, with clear instructions for translators about context, target audience, text lengths, naming conventions, and other technical requirements.

The number of tasks going into localizing a game grows exponentially with each new language added. Because of that, many companies will often outsource many of these tasks to studios and individuals who specialize in localization and are highly familiar with the requirements that different territories pose. As such, localization must be planned very carefully with ample time allocated for the process to fit in the game's production schedule.

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What are some common game audio tools?

Game audio uses a variety of tools, depending on the focus of the creator. Unfortunately, like everything in game audio, nothing is “one size fits all.” It’s a contextual decision, based on many factors. However, there are underlying rules on how to choose the right tool for the right job.

SOUND RECORDING

For recording sound effects, it is handy to have a mobile recorder and some microphones. To avoid diving too deep into the gear world, here’s a quick rundown: for outdoor recordings, you could use either a shotgun microphone to capture a sound source very precisely or an MS/XY setup (built into popular handy recorders like the [Zoom H1n](#) or [Sony PCM-A10](#)) for a wider stereo image. For both, you should consider using a [blimp/windscreen](#) to avoid recording only wind rumbling.

Additionally, a contact microphone or hydrophone can help you find unique textures for further sound design. (Check out Zach Poff’s “[Choosing a](#)

[Contact Mic For Field Recording](#)" and "[Choosing a Hydrophone For Field Recording](#)")

Using directional microphones [...] allows you to focus the point where you want to capture a sound.

Using directional microphones, like shotgun microphones, allows you to focus the point where you want to capture a sound. This makes it easier to record and edit a single sound source in a noisy outdoor environment. For musical recordings, you might look for small or large diaphragm microphones to capture the instrument(s) in detail.

There are also EMF (electromagnetic field) microphones (such as LOM's [Elektroucho Pro](#)) that allow you to capture electromagnetic sounds from electronic devices that are not usually heard by the human ear.

For optimal recording, you should invest some time and/or money to find an adequate environment in which to record.

For more gear advice, check out [field recordist George Vlad's list of recommended gear](#). if you want practical advice on how to do field recording, check out this [free technical field recording course by Ian Smith](#).

EQUIPMENT

Regarding computer specs, many people think you need a crazy gaming PC with 16 cores minimum to develop a video game. Luckily, computers have gotten so much stronger over the years that you can develop video games even on a simple laptop. In regards to sound design, you might want to consider getting an audio interface. Apple computers come with a built-in audio card that works pretty well, while PC users would have to either get an external card or download ASIO4ALL to avoid latency.

The key spec to focus on should be processing power, followed by memory, followed by graphics performance

Keep in mind that testing unfinished or unoptimized games in the engine for the integration part of game audio can be quite demanding. Generally, processing performance is paramount when it comes to audio creation so if you're considering building or buying a new computer, the key spec to focus on should be processing power, followed by memory, followed by graphics performance.

However, you don't need to spend a fortune on equipment from the get-go. It's possible to start with a [lean game audio setup that doesn't break the bank](#) and you can then expand if and when your needs change. Juan Uribe also has some great insights on [how much \(or how little\) gear and software you need to be a sound designer](#).

From a software perspective, there are many options available, each with great features and user bases. We will only point out some of the more commonly used ones:

DAW

For a digital audio workstation (DAW), [REAPER](#), created by Cockos Incorporated, has become increasingly present in the game audio industry over the past few years. REAPER is an open source DAW with daily community updates – although Avid's [Pro Tools](#) and Steinberg's [Nuendo](#) are still popular choices. The two major things to note are that REAPER has incredible customizability, and its render options for individual items, time frames, etc. are far more extensive than any other DAW. Especially for game audio, where you might need to render hundreds of audio assets from the same session, having them all rendered in one action and already named correctly saves a lot of time. Javier Zumer has created an in-depth guide on [how to make the most of REAPER for game audio](#).

Additionally, REAPER runs on Windows, Mac, and Linux, is extremely lightweight and efficient, and can be expanded almost infinitely thanks to its lively community of developers providing workflow-enhancing extension scripts. Chris Bolt has compiled an extensive list of [free tools and scripts](#)

[for REAPER](#) to enhance and speed up the audio workflow.

PLUGINS

DAWs come with a set of highly capable stock plugins right out of the box. When starting out, these plugins are crucial to get familiar with as they already provide you with a very rich feature set that will let you handle the most common audio tasks as well as some pretty involved processing. Other great free additions are the [MeldaProductions MFreeFXBundle](#), [Kilohearts Essentials](#), [AirWindows](#), and [Tokyo Dawn Labs](#) bundles. [Xfer's OTT multiband compressor](#) is also a staple in sound design and has been completely free ever since its introduction.

In some cases, the original source recording or synthesized sound might not need any plugin processing at all

The use of plugins can vary wildly depending on the type of project, the style of the sound designer, and the needs of the specific sound that's being made. In some cases, the original source recording or synthesized sound might not need any plugin processing at all. In other cases, like a very sci-fi or alien themed world, you might rely on long chains of plugins to shape the sound exactly how you want it.

What's important to remember is that there is no correlation between plugin usage and quality of sound. What matters the most is how the final sound feels, and the sound designer can find many different paths to get there.

For specific plugins, preferences vary. It's important to know what you want to achieve and how to get there. When it comes to plugins for different use cases, in this [game audio tools](#) highlight, Greg Lester highlights a number of free and paid options in selected categories: [Bundles](#) • [Analysis / Spatial](#) • [Distortion / Saturation](#) • [Delay](#) • [Dynamics](#) • [EQ](#) • [Modulation](#) • [Reverb](#) • [Synthesis](#) • [Stock plugins](#) • [Other](#)

GAME ENGINES

The two most common game engines are [Unreal Engine](#) and [Unity](#). Both provide built-in sound systems. If you plan on working in game audio and integrating sounds, it might be a good idea to at least familiarize yourself with these engines. Dave Raybould has compiled an extensive primer on how to [get started and make the most of game audio in Unreal Engine](#), and Bjørn Jacobsen's has a video series on [game audio in Unity](#).

Many larger game studios will develop their own proprietary engines in-house to allow for even tighter optimization and integration. These game engines are usually a closely guarded secret privy only to those working within these companies. Even if you might not be familiar with a specific engine, understanding the basics of game development translates across all engines and coding languages.

MIDDLEWARE

As previously mentioned, game audio professionals often use audio middleware – additional software that provides a better overview of the sounds and is overall more “audio-friendly” and easier to understand than raw code. Audio middleware takes over the role of an audio engine, which handles the playback of all sound effects, music, and dialogue in a game, conveniently stored in so-called soundbanks.

The two major competitors here are [Wwise](#) and [FMOD](#). FMOD's interface is a bit closer to a DAW and thus potentially easier for audio people to understand, while Wwise offers more in-depth options for the integration process and can scale into very large projects without losing overview. Tzvi Sherman has created a quick overview of the [audio opportunities available in Wwise](#).

A huge benefit of utilizing middleware in the development process is that it gives the audio team more independence

A huge benefit of utilizing middleware in the development process is that it

gives the audio team more independence as audio features can be tested and implemented without having to significantly alter the game's main engine project. Additionally, middleware provides a much larger feature set than any game engine's integrated audio tools as it is highly specialized and the result of many years worth of development and iteration.

Wwise and FMOD are built to allow you to create complex interactive sound and music systems, dynamic mixes, and utilize real-time processing via a graphical interface in a manner that is highly modular and tweakable. With these available capabilities, there's less need to bother your programmers, for which they'll be thankful.

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What are some good sources for game sound effects?

If you have the time, budget, and know-how to do so, it is always best to record your own sounds. Not only can you get exactly what you are looking for and avoid digging through sound libraries to find a compromise, but it is simply more fun to go out "in the wild" to gather your own source material.

Sound Designer David Filskov maintains an extensive list of ideas on how to [create your own sound effects](#). Nevertheless, sometimes recording is just not an option, and especially when starting out, having some libraries is a good starting point.

You want to look for libraries offering variations of the same sound, so you can easily create auditory diversity for your game

Generally speaking, there is not really a separation between game audio sound libraries and film sound libraries. However, something you might need to be aware of is that in games, the player might retrigger the same action over and over again because every game loop consists of repeating elements. This is why you want to look for libraries offering variations of the same sound, so you can easily create auditory diversity for your game. Sound libraries specifically targeting game audio often feature a number of sound effects in individual files, whereas film focused ones contain longer files with multiple variations within that file.

FINDING GAME SOUND EFFECTS

When it comes to actually finding sound effects, there are multiple ways of doing so. Sound effects libraries can be found on sites such as A Sound Effect, where independent sound creators releases targeted and often affordable sound libraries, providing an unrivalled coverage when it comes to the breadth of sound types available. Cloud sound effects libraries, such as Soundly, also provide access to a number of sounds, as well as offer librarian features. It requires you to pay an ongoing subscription fee to access the full set of features.

There are also larger sound library creators such as BOOM Library, Sound Ideas, SoundDogs (now a division of Sound Ideas), Pro Sound Effects and others that provide individual sound libraries (or single sound effects in the case of SoundDogs) in a number of genres, as well as bundles and subscription packages.

INDEPENDENT SOUND EFFECTS

One challenge with catalogs that have been available for many years – sometimes even decades – is the risk of the sounds being overused. The independent sound community has risen as a movement to counter that with [indie sound effects libraries](#). With individuals in the community releasing smaller, targeted, and affordable sound effects libraries in nearly every genre imaginable, they offer fresher sound effects that can be found anywhere else – available to buy as you need them. While the quality can vary between vendors, strong competition in the independent sound community marketplace and the establishment of metadata standards such as Tim Nielsen's [Universal Category System](#) has caused a general uplift in quality in independent releases. To give an idea of the scope of these independent sound libraries, A Sound Effect is home to the world's largest collection of independent sound effects libraries, with more than 7,000 unique sound effects libraries available from independent sound creators around the globe. And typically, 10-20 new sound libraries are released every week.

OTHER WAYS TO FIND SOUNDS

A new [resource for free sound effects is the A Sound Effect Cloud](#) – it features 1000+ free sounds from independent sound community members, as well as 100+ customizable music tracks. There are also a number of reputable free sound effects sites available such as Zapsplat – as well as sound effects swaps and general giveaways. With these it's crucial to check the licensing terms to ensure you can use the sounds for your project in the way you intended. When it comes to free sound effects sites, you may also want to do some investigating to see how long the given site has been around and what people are saying about it, as there have been a number of free sound sites that are simply giving away sounds taken without permission from commercial sound libraries.

AI sound effects generation – begin developed by companies such as

ElevenLabs and MyEdit – is evolving rapidly. Once again, be sure to read the usage terms to ensure they fit your project.

Game sound effects highlights:

Most popular game sound effects in the last 30 days:

Need more game sound effects?

[Browse the game audio sound effects category here](#)

Search for game sound effects:

If you're looking for specific game sound effects, try this custom sound effects search engine:

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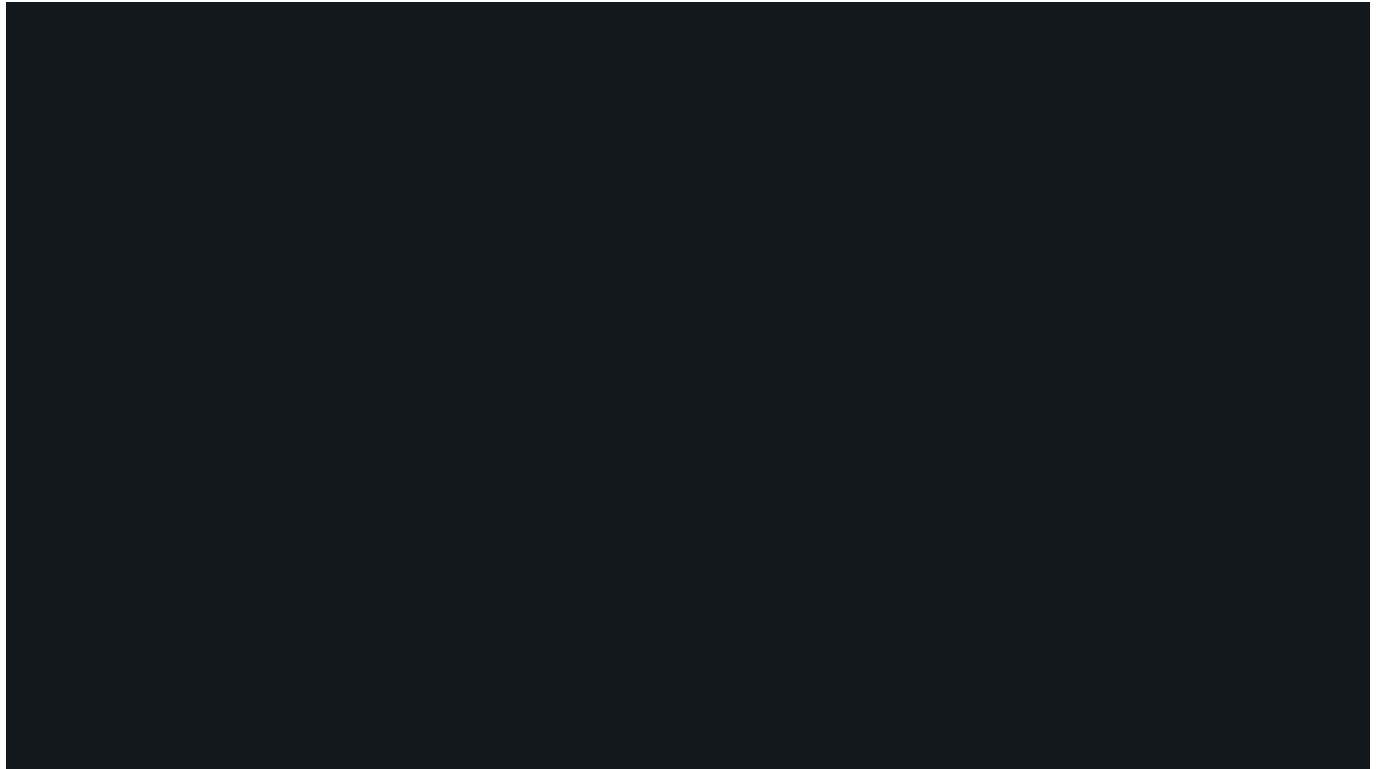
Is using sound effects libraries acceptable?

When learning sound design you might be tempted to try and record or synthesize every single piece of sonic source material yourself. However, with tight deadlines and the mountain of work that game audio professionals often face, it becomes clear that recording everything in a bespoke manner is not realistic and often too expensive. Especially when you're flying solo, using libraries is important and expected. Libraries contain valuable source material that you can transform in any way you want and they let you sonically travel to places you would've never gotten to otherwise.

Libraries contain valuable source material that [...] let you sonically travel to places you would've never gotten to otherwise

Say the game you're working on is set in the Amazon rainforest. It would be very hard to create a faithful soundscape of that area without traveling there and recording it yourself, were it not for libraries from recordists around the globe! All that is to say that sound effects libraries are an important tool for sound designers at all levels of their careers and are by no means a crutch or "cheating." If you are looking for specific sound effects, find out what the [independent sound community has to offer here](#).

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How to learn game audio – what's the best route for education in game audio?

LEARNING ONLINE

Want to teach yourself game audio? For some first impressions, there are a few different YouTube channels (e.g., [Akash Thakkar](#) and [Marshall McGee](#)) and podcasts like [The Tonebenders](#) who have a general focus on sound design, specifically about game audio content. The [Audio Podcast Alliance](#)

is a collection of different audio related podcasts that can serve as a great inspiration. [Quit Aspiring](#) by Adam T. Croft is a fantastic starting point and great read for anyone who is interested in game audio as a career. [The Game Audio Strategy Guide](#) by Gina Zdanowicz and Spencer Bambrick is a really helpful book that covers a wide range of game audio related topics.

'Quit Aspiring' by Adam T. Croft is a [...] great read for anyone who is interested in game audio as a career

Greg Lester runs a well-maintained resource page for [learning game audio](#). Additionally, Wwise has a well-documented [learning series](#) on their website, which – together with their demo game “The Cube” – allows you to get a good practical start with their middleware. There are many great blogs and resources found on social media, such as A Sound Effect’s dedicated [Game Audio blog category](#), and [Game Audio Power List](#), and even an audio-related social media site called [Airwiggles](#). Here you can find all sorts of helpful information for game audio and ask questions to the community.

If you want to take courses in game audio, the [School of Video Game Audio](#) offers online courses in Wwise, FMOD Studio, Unity, Pure Data & Unreal. You can also find courses at the [Game Audio Institute](#), as well as take Berklee Institute’s [game audio 101 course](#).

HANDS-ON EXPERIENCE

The best way to start is by learning tools like a game engine hands on. Even if many things might seem unclear at first, you’ll gain a lot of experience by getting involved in demo projects. Most publicly available game engines have tutorials and demo levels available on their websites, and there are many tutorials online on how to set them up with audio middleware and how to integrate your sounds – again, David Raybould’s [primer on game audio in Unreal Engine](#), [Sergio Ronchetti’s series on FMOD game audio skills](#), Audiokinetic’s [guide on getting started with Wwise](#), or Bjørn Jacobsen’s video series on [game audio in Unity](#) can be great starting

points. It's highly recommended that you check out tutorials and – even more importantly – try it out yourselves.

GAME JAMS

One of the easiest ways to gain tangible experience is through so-called Game Jams. These are events (either on-site or online) where developers, artists, audio designers, etc., come together to create a game based on a given topic within a limited time frame. The most notable game jam is the [Global Game Jam](#) held in January each year, but there are also many smaller jams hosted on platforms like [itch.io](#), where developers can release their own small (mini)games or prototypes.

If you have already gained some experience, you might consider doing an internship at a game studio or game audio company. Even though larger companies may have more well-known projects or clients, you should also consider smaller companies, as you might receive more responsibility there.

For continuous learning, the most important tip is probably to stay in contact with as many different game audio professionals as possible. If not at your workplace, open forums or online conferences can be a great opportunity for that as well (e.g., [Airwiggles](#) or [GamesCom](#), [Game Developers Conference \(GDC\)](#), etc.).

If you're just starting out, don't be overwhelmed by all this information. Again, since there are various fields in game audio, you might want to set your personal focus; just pick something that interests you right now and start with that.

A similar event that can help you gain a similar experience is the [48 Hour Film Project](#). Each film team is given a specific common element that all teams would strive to incorporate into their films, and the final film must be completed by the end of the weekend. Unlike the Global Game Jam, awards are presented at a future live awards ceremony and screening. This

event gives you the experience of working up against a deadline, and learning how to work with teams of different skills (technical and soft skills).

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How to get started and succeed in game audio – tools, tricks and ways to find jobs?

If you are just starting on your game audio journey, you will know that the current state of the industry has led to an increased scarcity in junior roles. Pair this with an overabundance of highly qualified and tenured talent having recently been laid off, and it's not hard to see that job postings are not the ideal place to start for aspiring juniors.

You'll have a much better chance of being recommended if you're known to be a great person to be around and have the chops to back it up

As with any creative job, networking and making friends in the industry is

critical. Many smaller studios or solo devs are unlikely to turn to job boards when looking for one or more audio professionals to help bring their vision to life. Instead, word of mouth is often paramount and you'll have a much better chance of being recommended if you're known to be a great person to be around and have the chops to back it up.

SHOWCASE YOUR WORK

Don't be afraid to put your work out there, no matter how early on in your career it may be. Sharing little bits and pieces of your process and what you're working on shows people that you're invested in the craft and love what you do. Passion is contagious, especially in such a passion-driven industry.

Make sure to have a place to send people to where they can learn more about you and your work. For starters, it's worth setting up your own domain and create a simple website that highlights your work. For discovery, you can also set up a portfolio on sites like [Soundlister](#), where you can showcase your work and include links to your other platforms, such as your own websites, LinkedIn profiles, YouTube channels etc.

To [improve your audio showreel](#), sound designer Lewis Thompson suggests focusing on elements like adding more details to ambiences, beefing up the Foley elements, experimenting with mono sound effects and more. If you're looking for advice on creating or improving your game audio work reel or demo reel, Senior Sound Designer Paul Boechler shares valuable tips, advice, and resources to help you make the right impression and progress in your audio career in his [How to Build a Strong Game Audio Portfolio](#) guide.

Another great way to make new friends and demonstrating your skills is by sharing knowledge and helping others. If you have some expertise with a certain plugin suite, DAW, or sound design technique, why not help people out who might be stuck? Places like [Airwiggles](#) or Discord groups are a great place to give back to the community.

NETWORK

Don't forget to network outside of the audio community. It's easy to get caught up just talking about audio things with audio people. But remember, other audio people are less likely to find themselves in a position where they can hire other audio people. Making friends and meaningful connections with people all across the different disciplines within game dev is important and will also help broaden your horizon.

Regardless of in-person or online, the most important thing when networking games is actually quite simple: just be nice and make friends. In general, people in game dev tend to be incredibly curious, welcoming, and eager to meet new peers. Simply saying "hi" and just chatting about games you've been playing, movies you love, or what you're passionate about can go a long way in making new friends and fostering connections that can eventually bloom into collaborations and sometimes even permanent jobs!

Be genuine, have fun, and help uplift the community and you'll see that people will love having you around. Don't think of every interaction as a potential transaction. Game devs are usually quick to notice when someone is just talking to them in hopes of landing a job. Or, as Production Director Kenn White puts it: [Want a job in the game industry? Stop networking](#) – and go find your people instead.

To find fellow game devs in your area, go on the hunt and search the web for any game dev meetups, clubs, or societies in your local area or nearest large city. Airwiggles maintains a [huge resource for local audio communities here](#), which could be a great starting point. There could be Discord or Slack channels where local devs mingle, or you might even find that there are regular monthly meetups. You may be surprised to find that an active game dev community might be closer to you than you think. Another great way to find out what's happening near you is to look for people from your area who are working in games and simply send them a message asking if they are part of any groups or local events.

FREELANCE WORK

Freelance work benefits from skills that all people who are self-employed need. This includes skills such as marketing, sales, and negotiation – not just audio related skills. Networking is an essential part of working as a freelancer – however you want to go about it. If you have a lot of patience and time, you can use it to cultivate connections online and on social media. Networking events can be helpful to find peers and connect with people in person.

Working as a freelancer in sound comes can be super rewarding, but comes with its own set of challenges

Working as a freelancer in sound comes can be super rewarding, but comes with its own set of challenges, such as how to create financial stability when you're living from project to project, how to cope with the ups and downs of the creative industries, how to strike a healthy work/life balance, and more. Melissa Pons has some great insights on how to live the [audio freelancer life](#).

Pricing is another challenge. Should you, for example, [work for free on some projects](#)? And how do you set [sustainable and competitive rates for your audio work](#)? Often, there could be a [better alternative to outright working for free](#), to make your creative work sustainable, manage customer expectations, avoid devaluing your work and undercutting other creatives.

GETTING FEEDBACK

Getting regular feedback on your work is invaluable. That's why finding someone you trust to give you feedback is crucial. You can look for game audio mentorship on [Game Audio Learning](#).

Simply and humbly ask them for feedback on something you've been working on

Another point to consider: one of the easiest ways to **not** land a job is to go around messaging people asking to get hired. A much better way to make established professionals aware of you is to simply and humbly ask them for feedback on something you've been working on. You'll be surprised at how happy and eager to help even very established professionals can be if you just ask nicely.

FINDING AUDIO JOB OPENINGS

When looking for job postings, the [Soundlister Audio Jobs](#) page is a fantastic starting point for new and open roles for game audio professionals. You can get weekly audio job highlights – typically featuring around 20 new opportunities – sent straight to your inbox by [subscribing to the Soundlister Audio Jobs newsletter here](#).

Keeping your website, resume, and demo reel current are practical and useful steps that you can do to maximize your chances of getting an interview. Knowing what you want to do is a powerful ally in the search for your perfect role. You might consider tailoring your application to the ideal studio and position you want to work in. Similarly, consider tailoring your reel to the kind of studio you'd love to work for and the type of games you'd like to work on.

BRANCHING OUT FOR ADDITIONAL REVENUE

One way to make a life in sound sustainable is to take on multiple types of audio jobs. So, if there's a downturn in one part of the creative industry, you can land projects in other areas that aren't affected. If you're looking for inspiration on how to expand your creative toolbox, the [Sound Success Guide](#) is a free 60+ page guide with insights from 20 industry experts on how to get started and succeed in 18 different types of audio jobs:

[Sound Success Guide](#)

BOUNCING BACK FROM A LAYOFF

The game industry can be a volatile one, with occasional rounds of mass-layoffs that can affect people in game audio as well. In an [in-depth guide on how to cope with layoffs in game audio](#), game audio veteran Damian Kastbauer suggests that the key to getting back on your feet after layoff actually starts while you're still employed, by building and maintaining a strong network and continuously staying updated on new technologies in game audio.

The key to getting back on your feet after layoff actually starts while you're still employed

If you're affected by a layoff, activate your network of friends, family, and colleagues for support, outreach and referrals, take care of practicalities such as unemployment benefits, and start looking for new opportunities.

In his [guide to help you power through a layoff](#), game audio veteran Brian Schmidt echoes the approaches suggested above, and also highlights the importance of continuously making yourself visible in the community by taking part in industry events and generally sharing your expertise.

Bonus: Get some insights on how to get hired – from your potential employee's perspective – [here](#).

FOSTERING CREATIVITY AND PREVENTING BURNOUT

Having to be creative on demand can be taxing on your mental health, and whether you're a game audio freelancer or working in-house, it will be intense and stressful work at times – with crunch periods, tight deadlines and sometimes insecure job situations. That's why it's crucial to do what you can to keep your creative fire burning, look after your mental health and to protect yourself from burning out. How do you do that? In [this interview on fostering creativity and preventing burnout](#), veteran composer Chance Thomas highlights the importance of setting boundaries,

scheduling creative work for when you're most productive, prioritizing non-work activities and more.

To [set your creativity free and remove inhibitions](#), veteran audio director and sound designer Mark Kilborn recommends – and provides ideas for – practicing breaking the “rules” when it comes to approaching a creative challenge, forcing yourself to go outside of your creative comfort zone, acknowledging and managing your inner critic, fighting your fear of failure, using meditation to relieve stress, and other approaches to stay sane and creative in a demanding environment. Juan Uribe highlights that it's important to acknowledge that, given the huge scope of game audio, it's [easy to get overwhelmed with how much there is to \(potentially\) learn](#), and he provides ideas on how to deal with that, and to help you focus.

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We hope you found this guide on everything game audio useful! Let us know if you have any comments – and once again, a big thanks to Chris Bolte, Bonnie Bogovich, Rob Bridgett, Joshua Hank, Juan Uribe, Yannick Winter, and Shirly Spikes for their insights. Please share this guide in your network so more people can benefit from it:

Please share this:

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